WISCONSIN BREAST CANCER DETECTION USING L1 LOGISTIC REGRESSION

2023 International Conference on Intelligent Systems for Communication, IoT and Security (ICISCoIS) | 979-8-3503-3583-5/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICISCOIS56541.2023.10100387

Dr.R.Rekha,
Associate Professor,Department of Information Technology,
PSG College of Technology,
Coimbatore, Tamil Nadu, India.
rra.it@psgtech.ac.in

K.L.Vinoci,
1st Year M.Tech ,Department of Information Technology,
PSG College of Technology,
Coimbatore, Tamil Nadu, India.
22ma01@psgtech.ac.in

***Abstract*—The detection of breast cancer type from the Wisconsin CSV dataset is described in this paper. One of the most frequent malignancies in women is breast cancer. An earlier diagnosis will result in the cancer being completely cured. However, this cancer's development may endanger life. The logistic regression model with lasso regularization and the logistic regression model without lasso regularization were contrasted. As a result, for this Wisconsin dataset, the performance of logistic regression with lasso regularization outperforms that of the conventional logistic regression model as well as the ridge logistic regression model.**

***Keywords* —Logistic Regression, Lasso Regularization, Wisconsin Breast Cancer Dataset**

I. INTRODUCTION

Cancer is nothing more than the accelerated, abnormal development of cells in a limited area. The development of cancer can occur for a number of causes, including genetic abnormalities, viral exposure, family history, and alterations in lifestyle. The human body contains cancer in a variety of locations. According to statistics, women are more likely to be diagnosed with ovarian and breast cancer. The incidence of breast cancer has surpassed the prevalence of cervical cancer in Indian women, with 1,78,000 new cases being detected each year. Men typically receive diagnoses for prostate, colon, lung, and skin cancers. According to reports, for every 4 minutes a women in India get diagnosed with breast cancer. Cancers are typically divided into two categories: primary cancers and secondary cancers. Cancers that are considered primary remain in one area of the body and do not spread to other organs. Secondary cancers, however, pose a greater risk to our lives since they have the potential to spread to other parts of our bodies. Additionally, the primary cancers are divided into benign and malignant cancers. The most advanced stages of cancer make survival more challenging, and more than 50% of Indian women have advanced breast cancer. According to reports, Indian women had a 60% post-cancer survival rate, compared to 80% in the US. Cancers that are benign are entirely treatable and not fatal. The lumps are typically used to diagnose them. If malignant tumors progress to their terminal stages, they pose a threat to life. Early discovery will aid in early diagnosis, which will lead to full recovery. Symptoms of the breast cancer across women are a breast or underarm lump which is the initial sign won't go away. Breast cancer can be even swelling in armpit or collarbone. This may be a sign that breast cancer has spread to neighboring lymph nodes. Although lumps normally don't hurt, there may be some pain and tenderness. Change in the shape of breast isalso considered to be a sign of breast cancer. However the detection of breast cancer is usually done by performing mammography. Therefore, it's crucial to find breast cancer as soon as feasible. A machine-learning model assists in achieving this.

II. LITERATURE REVIEW

Shruthi Rao et al. [2] explored how preprocessing techniques impact model accuracy in their paper. Preprocessing is a crucial stage as it can change all of the missing values in the dataset. Since the data was gathered in various locations, there is a possibility of missing values within the dataset. Even if the datasets have been preprocessed, they must still be standardised or normalised because the model may be biassed towards high-valued features. Thus, Dalwinder Singh et al. [1] and Rekha et al.[14] have explained the significance of data normalization and its impact on model accuracy. Following completion of preprocessing and normalization, it is critical for the model to have separate training and testing datasets, thus the dataset split. However, the model's efficiency is also affected by the number of features. It is not necessary for the model to use all of the features in the dataset. It is possible for a dataset to contain redundant and unnecessary features. With more features, training the model becomes more complex, reducing its efficiency. Rekha et al. [13] has discussed how feature selection is important in reducing the features. Thomas Rincy et al. [3] discussed how feature selection methods in machine learning affected model performance. There are numerous methods for selecting features from a dataset. The methods are classified based on whether the problem is supervised or unsupervised. As this Wisconsin dataset is supervised, the feature selection methods are filter, wrapper, and embedded. P.Saha et al. has explained about how forward selection approach is used for extraction of feature from the dataset. V.Pesala et al. [10] has explained about the procedure of backward elimination in feature selection. A.Nugroho et al.[12] had compared forward feature selection and backward elimination method from which the accuracy of backward elimination is better compared to the forward feature selection method. Y.Ono et al. [11] has discussed how recursive feature elimination method helps in feature selection. Shamsuddeen Muhammed Abubakar et al. [4] compared three different feature selection methods. However, when compared to the other methods, embedded

Authorized licensed use limited to: Saxion Hogescholen. Downloaded on March 12,2026 at 13:28:27 UTC from IEEE Xplore. Restrictions apply.

methods perform better for feature selection. Regularization is the process of adding bias to a model inorder to train and generalise it, with the parameters optimised using a loss function and optimizers. However, Johnson Kolluri et al. [5] explain that the L1 regularisation, also known as the Lasso regularisation, outperforms ridge regularisation for feature selection. Following the feature selection process, the selected limited features are used for machine classification. This isdone with the help of the classification algorithm logistic regression. Bristi Rani Roy et al [6] compared the Naive Bayes (NB), Support Vector Machine (SVM), Logistic Regression (LR), and K-Nearest Neighbor (KNN) algorithms for the Wisconsin breast cancer dataset and found that the Logistic Regression algorithm performed better than the other algorithms. However, no feature selection method was used on the datasets. F.Huang et al.[8] has explained about ridge regression, a type of regularization technique could be used in the dataset as it can give more generalization to the model. Jabeen Sultana and Abdul Khader Jilani [7] compared different classifiers for the same Wisconsin breast cancer dataset and found that logistic regression with ridge regression performs better than the other classifiers.

The main objective of the proposed model is

- To preprocess the given dataset by performing normalization feature scaling
- To perform feature selection with the help of Lasso regularization
- To classify the target attribute with the help of logistic regression

III. PROPOSED MODEL

The proposed model would usually consist of four phases. They are,

A. DATASET SELECTION

The dataset was obtained from the open dataset on Kaggle. The Wisconsin dataset would consist of 569 examples aggregated together in CSV data format. This dataset would have 32 features. The diagnosis attribute is taken into account and used as the target attribute.

B. PREPROCESSING

Data that has been extracted from various domains may not be consistent, so preprocessing is considered to be an essential step. Preprocessing is performed to ensure the accuracy of the data. If the dataset contains missing values, they can be manually replaced or replaced with mean using the SimpleImputer library function, or the values can be dropped from the dataset. However, since this takes less time and maintains data consistency, the SimpleImputer method could be used. Two methods can be used to avoid the model being biassed by the different range of values for different features. They are standardization and normalization, respectively. Standardization will aid in converting values within a range and may be useful in re- moving outliers. However, after performing normalization, the shape or distribution of data will change. Normalization could not be more effective with outliers. As a result, standardization is preferred for this dataset. However, after standardization, the target variable must be determined, so the dataset must be divided into dependent and independent variables.

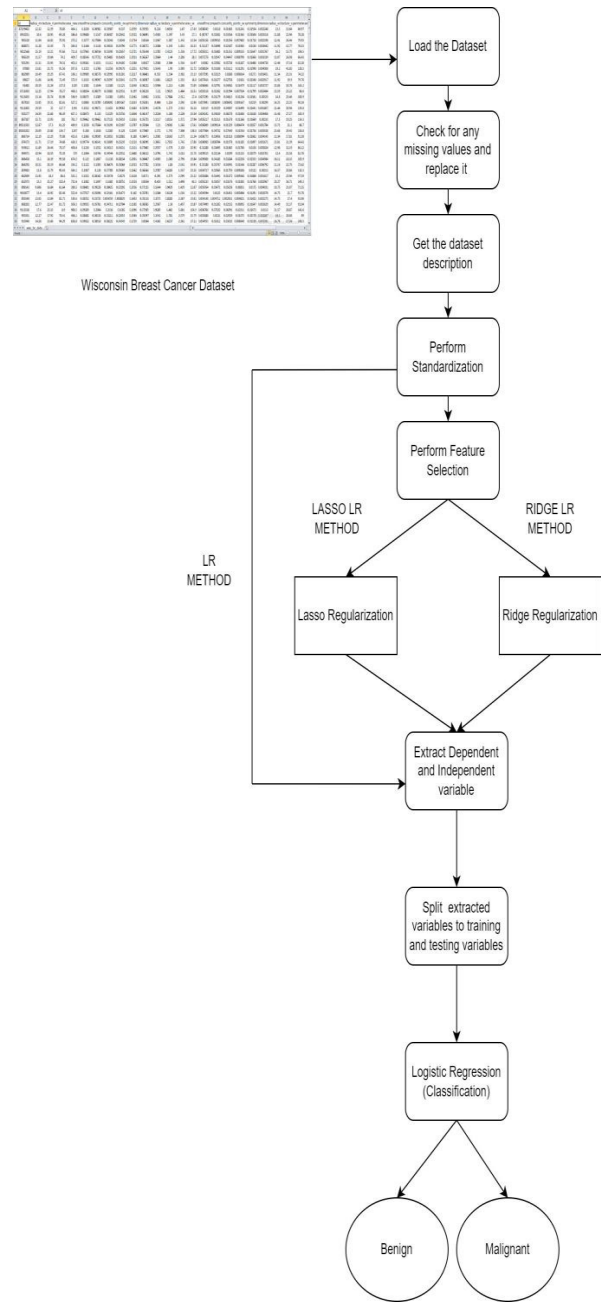

Fig. 3. 1. Image showing the workflow of the proposed model

The dependent variables are those that are used to calculate the target variable. Independent variables are the characteristics that can be used to select the target variable. Encoding the dependent variable is important because the target variable may be categorical and thus must be converted to a numerical

Authorized licensed use limited to: Saxion Hogescholen. Downloaded on March 12,2026 at 13:28:27 UTC from IEEE Xplore. Restrictions apply.

value. As this dataset contains binary classification, the target variable is encoded using LabelEncoder. The target variable is encoded and then passed along with the independent variable for training and testing dataset split. The dataset has completed its preprocessing phase after an individual split of the training and testing datasets for both independent and dependent variables in the ratio of 80:20.

C. FEATURE SELECTION

The main phase of the machine learning model is feature selection. It aids in the removal of redundant features, allowing the model to be trained with the fewest features possible and producing better results. When supervised feature selection methods such as filter, wrapper, and embedded are compared, the latter performs better than the others. Filter methods are those that use a statisticalparameter to select features. Some filter methods, such as the Pearson correlation coefficient, will consider the correlation of each independent attribute with those of the dependent attribute, and if a feature has a correlation greater than a certain threshold level, it will be chosen. A correlation test would be performed among the selected features, and the redundant features would be eliminated at the end. When considering the wrapper method, it would proceed in a recursive or step-by-step manner.

The forward selection approach, backward elimination approach, and recursive feature elimination are the three different algorithms contained within the wrapper method. The forward selection approach would start with no features and progress to the next stage by taking one feature at a time. It moves on to the next feature after determining the best feature in the current iteration by adding another feature to it. After iterating the required number of times, it would arrive at the best combination of features, and thus thefeatures could be selected. In the case of the wrapper method, in addition to those feature selection algorithms, it would use the filter method internally, as well as supervised algorithms to iterate at each step. Backward elimination methods would function differently than forward selection methods. At the start, all of the features would be present, and after each iteration, one feature would be removed from the feature set, and after the required number of feature iterations, the features could be selected. The recursive feature elimination method employs a ranking procedure in which a ranking is assigned to each feature and the number of features required is determined based on that. Overfitting is one of these methods' drawbacks. Overfitting can occur due to the limited number of datasets, causing the model to memorize well on the training dataset but fail to produce results on the testing dataset. A combination of filter and wrapper methods is used to address this flaw.

Regularization works by adding bias to the function, reducing variance and avoiding overfitting by decreasing generalization error rather than training error. Although there are three regularization methods, the Least Absolute Shrinkage and Selection Operation (LASSO) method is preferred for feature selection. A penalty term will be added to the model as it possesses high variance. The main distinction between L1 and L2 regularization is that Lasso can bring coefficients to zero by adding penalties,whereas L2 can only bring them closer to zero but not equal to zero. This would aid in the elimination of unnecessary features.

D. CLASSIFICATION

A statistical method that machine learning has adopted is called logistic regression (LR). Data scientists frequently utilize the LR approach to address classification issues. It is essential, herculean, and also simple to apply. In biological science, it is more frequently employed. When the target data is categorical—that is, when we must determine if a tumor is malignant or not—Logistic Regression is very liberally employed. Logistic regression is an extension of linear regression for resolving this issue because linear regression falters for classification models even though it works very well for regression models. A continuous numeric value is the outcome of linear regression. When a sigmoid function is used in logistic regression, the output is mapped to two or more discrete sets of classes. A logistic function is also used to predict the class. Both the classification model and the regression model exhibit excellent performance.

IV. PERFORMANCE MEASURE

A. CONFUSION MATRIX

A confusion matrix, additionally referred to as a contingency table or error matrix, is employed to indicate however well a classifier is playing. The rows of the matrix represent occurrences of the particular category, and therefore the columns of the matrix represent samples of the anticipated categories.

With the help of values in confusion matrix the performance metrics like accuracy, precision, recall, F1 score, MCC are computed.

B. ACCURACY

$$Accuracy = (TN + TP)/(TN + TP + FN + FP)$$

C. PRECISION

$$Precision = (TP)/(TP + FP)$$

D. RECALL

$$Recall = (TP)/(TP + FN)$$

E. F1 SCORE

$$F1\ score = 2 * (precision * recall)/(precision + recall)$$

F. MCC

$$MCC = (\mathrm{TP} * \mathrm{TN} - \mathrm{FP} * \mathrm{FN}) / \sqrt{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}$$

G. AUC

The Area under Curve separates the "signal" from the "noise" by plotting the TPR against the FPR at different threshold values.

Authorized licensed use limited to: Saxion Hogescholen. Downloaded on March 12,2026 at 13:28:27 UTC from IEEE Xplore. Restrictions apply.

MEASURE	WITH LASSO LR	WITH LR	WITH RIDGE LR
ACCURACY	99.1	98.2	98.2
PRECISION	100.0	97.3	97.3
RECALL	97.3	97.3	97.3
F1 SCORE	98.6	97.3	97.3
MCC	98.0	95.9	95.9
AUC	98.6	97.9	97.9

Table 1: Comparative table depicting the performance metrics of the three models

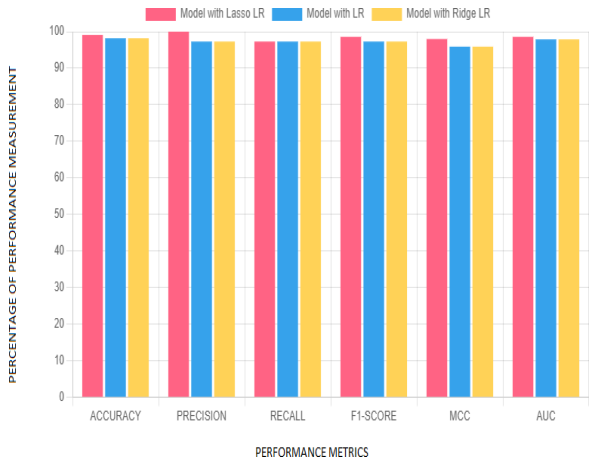

Fig.4.1. Graph depicting a comparison on the performance measure of the values

Since accuracy can sometimes produce incorrect predictions, F1 score comparison is also used to obtain an accurate result.

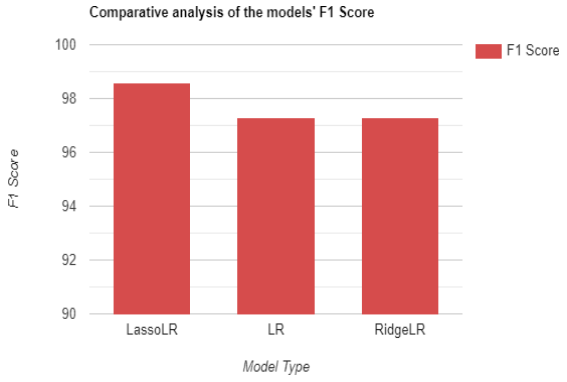

Fig.4.2 Comparative analysis of the model's F1 Score

Since the F1 Score and accuracy of the proposed model is higher than the existing model, it is evident that the proposed model outperforms the Ridge LR.

V. CONCLUSION

One of the most frequent cancers in women is breast cancer. But early detection and treatment are essential if it is to be completely cured. Therefore, in order to perform well in detection, the deep learning model must be accurate. As a result, the graph and table previously described can demonstrate how this proposed model (feature selection with Lasso coupled with Logistic Regression) could outperform the current model (only classification with Logistic Regression). However comparing the results of the proposed model with the Ridge LR, it clearly indicates that the accuracy of the Lasso LR is higher than the Ridge LR and LR.

REFERENCES

[1] Dalwinder Singh, Birmohan Singh, "Investigating the impact of data normalization on classification performance," Applied Soft Computing, Volume 97, Part B, 2020, 105524, ISSN 1568-4946, https://doi.org/10.1016/j.asoc.2019.105524.

[2] Rao, S., Poojary, P., Somaiya, J., & Mahajan, P. (2020). "A Comparative Study between various Preprocessing Techniques for Machine Learning.

[3] T. R. N and R. Gupta, "Feature Selection Techniques and its Importance in Machine Learning: A Sur- vey," 2020 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS), 2020, pp. 1-6, doi: 10.1109/SCEECS48394.2020.189.

[4] Abubakar, Shamsuddeen Muhammad & Sufyanu, Zahraddeen & Ibrahim, Abdullahi & Miyim, Abubakar & Arputharaj, Vijay & Kumar, Senthil. (2022). Comparisons of Filter, Wrapper and Embedded-Based Feature Selection Techniques for Consistency of Software Metrics Analysis. SLU Journal of Science and Technology. Vol4. 188-204. 10.56471/slujst.v4i.238.

[5] J. Kolluri, V. K. Kotte, M. S. B. Phridviraj and S. Razia, "Reducing Overfitting Problem in Ma- chine Learning Using Novel L1/4 Regularization Method,"2020 4th International Conference on Trends in Electronics and Informatics (ICOEI)(48184), 2020, pp. 934-938, doi: 10.1109/ICOEI48184.2020.9142992.

[6] B. R. Roy, M. Pal, S. Das and A. Huq, "Comparative Study of Machine Learning Approaches on Diagnosing Breast Cancer for Two Different Dataset," 2020 2nd International Conference on Advanced Information and Communication Technology (ICAICT), 2020, pp. 29-34, doi: 10.1109/ICAICT51780.2020.9333507.

[7] J. Sultana, "Predicting Breast Cancer using Logistic Regression and Multi-Class Classifiers," International Journal of Engineering and Technology, 2018.

[8] F. Huang, D. Li, J. Xu, Y. Wu, Y. Xing and Z. Yang, "Ridge Regression Based on Gradient Descent Method with Memory Dependent Derivative," 2020 IEEE 11th International Conference on Software Engineering and Service Science (ICSESS), 2020, pp. 463-467, doi: 10.1109/ICSESS49938.2020.9237632.

[9] P. Saha, S. Patikar and S. Neogy, "A Correlation - Sequential Forward Selection Based Feature Selection Method for Healthcare Data Analysis," 2020 IEEE International Conference on Computing, Power and Communication Technologies (GUCON), 2020, pp. 69-72, doi: 10.1109/GUCON48875.2020.9231205.

[10] V. Pesala, A. K. Kalakanti, T. Paul, K. Ueno, A. Kesarwani and H. G. S. P. Bugata, "Incremental Learning of SVM Using Backward Elimination and Forward Selection of Support Vectors," 2019 International Conference on Applied Machine Learning (ICAML), 2019, pp. 9-14, doi: 10.1109/ICAML48257.2019.00010.

[11] Y. Ono and Y. Mitani, "Effect of the random forests with recursive feature elimination for breast cancer classification," 2021 6th International Conference on Intelligent Informatics and Biomedical Sciences (ICIIBMS), 2021, pp. 95-96, doi: 10.1109/ICIIBMS52876.2021.9651590.

[12] A. Nugroho, A. Z. Fanani and G. F. Shidik, "Evaluation of Feature Selection Using Wrapper For Numeric Dataset With Random Forest Algorithm," 2021 International Seminar on Application for Technology of Information and Communication (iSemantic), 2021, pp. 179-183, doi: 10.1109/iSemantic52711.2021.9573249.

[13] Rajagopal, R. and Ranganathan, V. (2016) Critical Evaluation of Linear Dimensionality Reduction Techniques for Cardiac Arrhythmia Classification. *Circuits and Systems*, **7**, 2603-2612. doi: 10.4236/cs.2016.79225.

[14] Rajagopal R, Ranganathan V. Design of a hybrid model for cardiac arrhythmia classification based on Daubechies wavelet transform. Adv Clin Exp Med. 2018 Jun; 27(6):727-734. doi: 10.17219/acem/68982. PMID: 29877638.

Authorized licensed use limited to: Saxion Hogescholen. Downloaded on March 12,2026 at 13:28:27 UTC from IEEE Xplore. Restrictions apply.